Process Reward Model as Cross-Task Learner

An Extension of Vision-Language Model to Zero-Shot Medical Reasoning

Jiahua Zhang[1] · Yidong Tian[1] · Jinghao Liang[2] · Dianhan Lin[3] · Yiwen Cai[4] · Yuanqing Liu[4] · Zishan Huang[2] · Jingchun Ni[4] · Jianxing He[2,*]

[1]Nanyang Technological University, Singapore [2]Guangzhou Medical University, Guangzhou, China [3]Shantou University Medical College, Shantou, China
[4]Guangdong Medical University, Dongguan, China [*]Corresponding author: jianxinghe@gzhmu.edu.cn

MedTRACE | MM '26 · Rio de Janeiro, Brazil

MedTRACE
code & appendix

Motivation

CT volumes and gigapixel whole-slide images, the primary evidence for diagnosis, hold **hundreds of frames**; a VLM reads only a few, and new tasks emerge faster than labelled data.

- **Evidence overload.** Only a few frames are decisive; selecting them still relies on expert lesion annotations and per-task fine-tuning.
- **Scarce reasoning supervision.** Step-level chain-of-thought (CoT) labels are expensive, and Monte Carlo signals alone cannot tell whether a step is grounded in the image.
- **Modality-specific training.** Fine-tuning the VLM or the reward model for every new modality is prohibitive when labels are scarce.

MedTRACE trains a lightweight **process reward model (PRM)**, which scores every reasoning step rather than only the final answer, *once*. Deployment is a **training-free, zero-shot** pipeline: a frozen VLM handles unseen tasks and modalities with **no gradient updates**.

Method: Train the PRM Once, Infer Zero-Shot

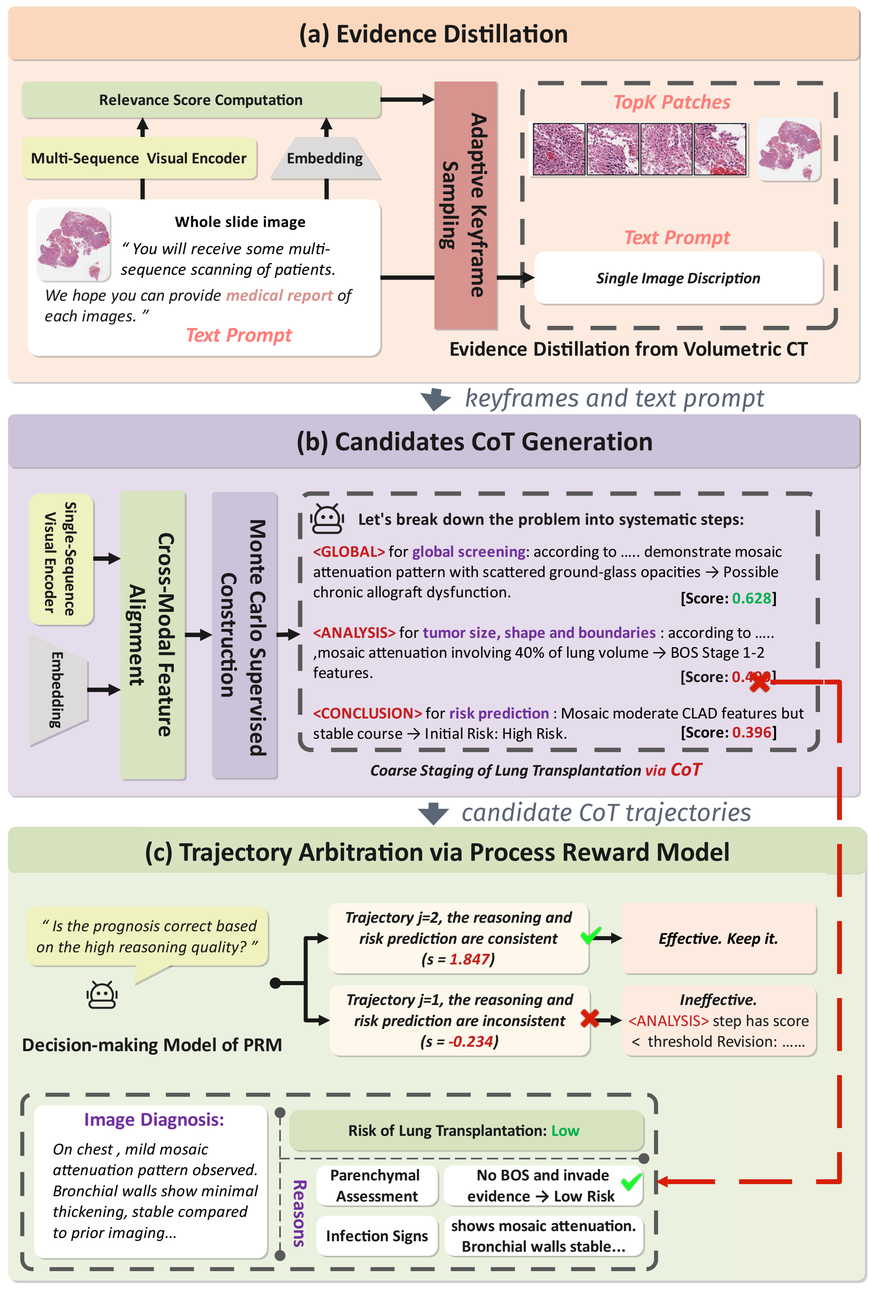
Offline: train the PRM once
source domains, labels available

- **Adaptive keyframe sampling** keeps the relevant frames; the frozen VLM samples diverse candidate reasoning chains.
- **Fused step-level supervision** labels every step automatically: text–vector matching checks *visual grounding*, Monte Carlo rollouts check whether the step still leads to the *correct answer*.
- No expert step labels: the PRM learns to judge each reasoning step.

Deployment: zero-shot
unseen task or modality, everything frozen

1. **Evidence distillation:** optimal transport aligns frame features with the text-instruction space and keeps the most informative keyframes, with no annotation and no training.
2. **Candidate CoT:** the frozen VLM writes several step-by-step reasoning trajectories.
3. **Trajectory arbitration:** the PRM scores every step, aggregates the step scores, and keeps the best trajectory.

Pipeline Overview

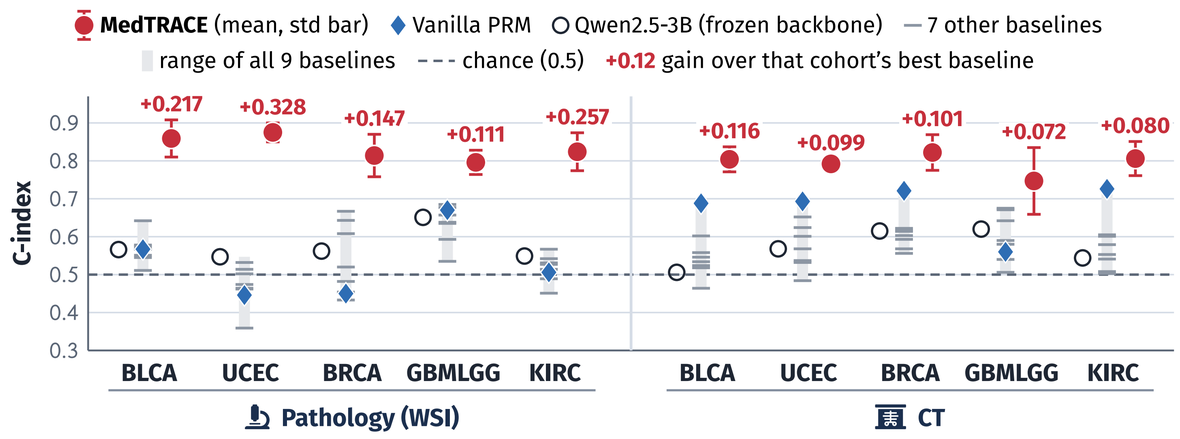


Results: Zero-Shot Cross-Task Transfer

0.834 — WSI survival: mean C-index — +0.247 vs. best baseline

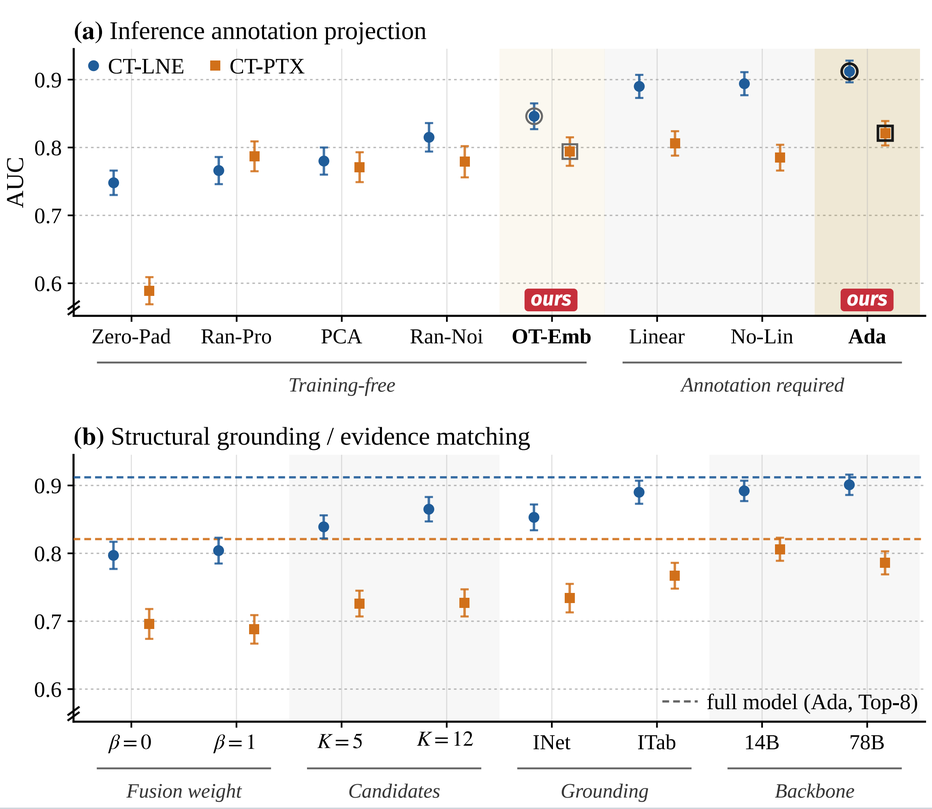
0.794 — CT survival: mean C-index — +0.117 vs. best baseline

10/10 — cohorts best — zero-shot, no TCGA training



Task: survival risk ranking on 5 public TCGA cohorts; a C-index of 0.5 is chance.
Protocol: PRM trained on private lung-cancer WSI & CT; zero-shot, no TCGA data seen in training.
Test-time-scaling baselines (Self-consistency, ORM, Vanilla PRM) also use Qwen2.5-3B.
Other baselines: llava-med-7b, RadFM, Mantis, LLaMA3-8B, LLaMA3-70B.

Component Ablation



AUC, private lung CT
LNE: lymph-node enlargement, 103 cases
PTX: pneumothorax, 172 cases

- **OT-Emb**, used at deployment: label-free, near the trained Linear.
- **Ada** is best but needs biomarker queries.
- **Fused labels** beat either signal alone (left pair).
- **Pool size:** 12 candidates beat 5.
- **ITab** beats **INet** grounding.
- **14B / 78B PRM:** far costlier, no clear gain.

Inference & PRM in Action

Zero-shot inference
unseen task · every module frozen · no gradient updates

1. **Evidence distillation** — keeps the slices most relevant to the question
2. **Candidate CoT** — frozen VLM writes 8 candidate reasoning chains
3. **Trajectory arbitration** — scores every step, keeps the best chain

PRM step scoring on a real CT case
stage 3 up close



CASE STUDY · LUNG CT · PATIENT SURVIVAL 11.3 MONTHS

Two sub-questions, one 5-step chain each, scored separately, not against each other.

Chain score: sum of step log-odds (above 0.5 adds, below subtracts).

0.7+ | 0.5–0.7 | near 0.5 | below 0.45

red text: contradicted claim green text: consistent finding

A Indicate the pleural contact

		0.5	score	log-odds
1	**Screening:** no pleural effusion, thickening or chest-wall mass		0.628	+0.524
2	**Evidence:** focal pleural contact with subtle retraction		0.575	+0.302
3	**Analysis:** chest-wall structures intact, no invasion		0.598	+0.397
4	**Synthesis:** lesion abuts the pleura; imaging criteria not met		0.748	+1.088
5	**Conclusion:** pleural contact, no radiologic invasion → No		0.812	+1.463

Final answer: **No** ✓ (correct) — chain score **+3.774**
no flagged claims

B Indicate the tumor involvement

		0.5	score	log-odds
1	**Screening:** right lung shows no clear focal mass or nodule		0.676	+0.735
2	**Evidence:** left upper lobe; conspicuous air-containing lesion	⚠	0.499	−0.004
3	**Analysis:** irregular solid mass, bronchial cut-off, no cavity		0.561	+0.245
4	**Synthesis:** thick-walled cavitary lesion, contents mostly air	⚠	0.397	−0.418
5	**Conclusion:** persistent thick-walled solid mass → Yes		0.396	−0.422

Final answer: **Yes** ✓ (correct) — chain score **+0.137**
steps 2 and 4 contradict the solid mass of step 3

Right answer, flawed chain: steps 2 and 4 describe an air-containing lesion that contradicts the solid mass of step 3. The PRM is neutral at step 2 and penalises steps 4 and 5, so the chain scores near zero, a flaw an outcome reward would miss.